

Name: Abraham Abebe
Course: Cybersecurity
Date: July 29, 2026
Title: Social Engineering Toolkit (SET) and Reconnaissance.

Exercise 4: OSINT Reconnaissance with Maltego

Objective

The objective of this exercise was to use Maltego Community Edition (CE) to perform Open-Source Intelligence (OSINT) reconnaissance by collecting and visualizing publicly available information. The exercise demonstrates how relationships between domains, email addresses, individuals, and other entities can be mapped to improve understanding of an organization's digital footprint.

Environment

Item	Description
Host Operating System	Arch Linux (Main Boot OS)
Tool Used	Maltego Community Edition (CE)
Java Runtime	OpenJDK
Purpose	Open-Source Intelligence (OSINT) Reconnaissance

Procedure

Step 1: Launch Maltego Community Edition and Create a New

A new graph was created to begin the reconnaissance exercise.

Step 2: Add the Target Domain and Execute Maltego Transforms.

A **Domain** entity representing the target organization was added to the graph.

Example Domain

`example-corp.local`

Maltego transforms were executed on the domain entity to gather publicly available information.

The transforms attempted to identify:

- DNS records
- Email addresses
- Domains
- IP addresses

- Related entities

Observation

Maltego collected available OSINT information and displayed the discovered entities as connected nodes within the graph.

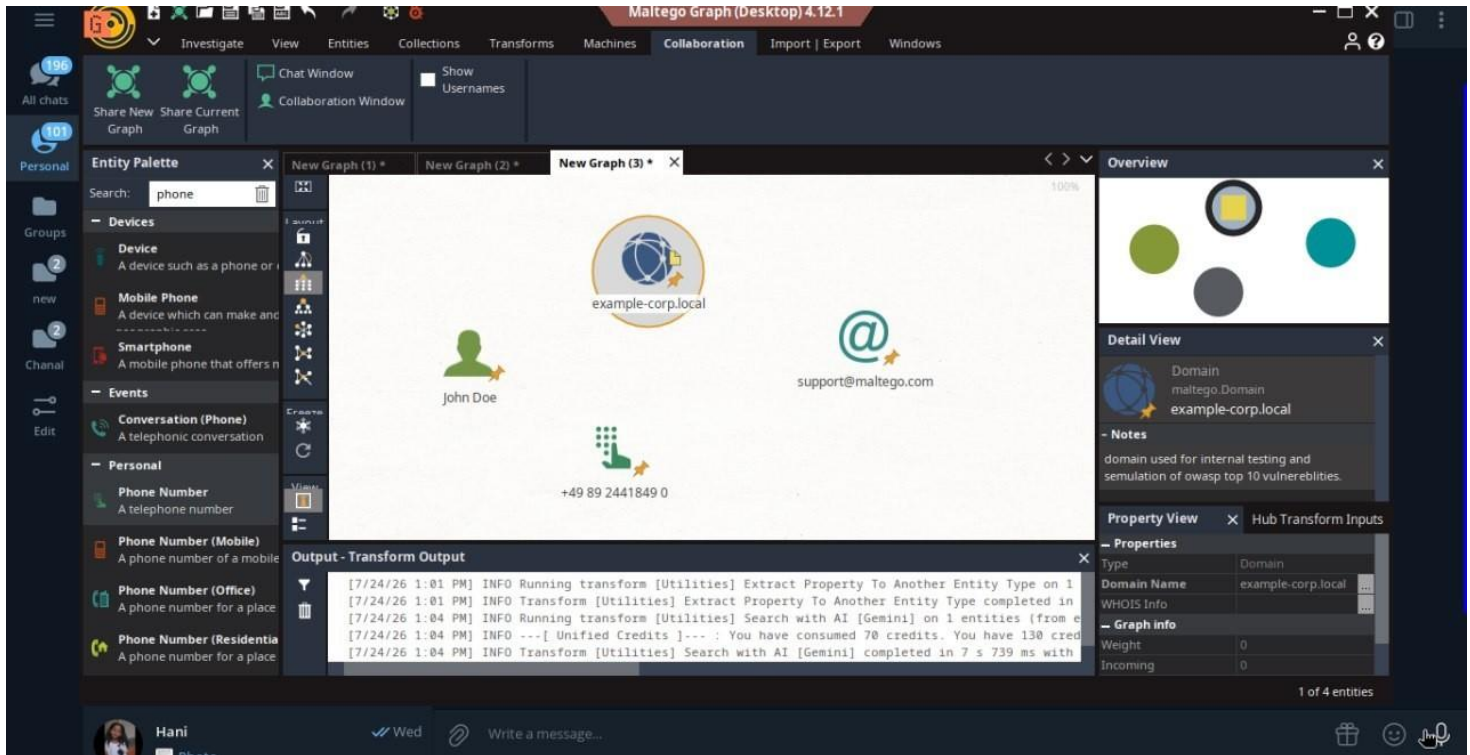


Figure 1: Maltego Transform Results.

Step 3: Build Relationship Graphs

Additional entities such as **Person**, **Email Address**, and **Phone Number** were added and linked where appropriate.

Observation

The graph visually represented relationships between the discovered entities, making organizational connections easier to understand.

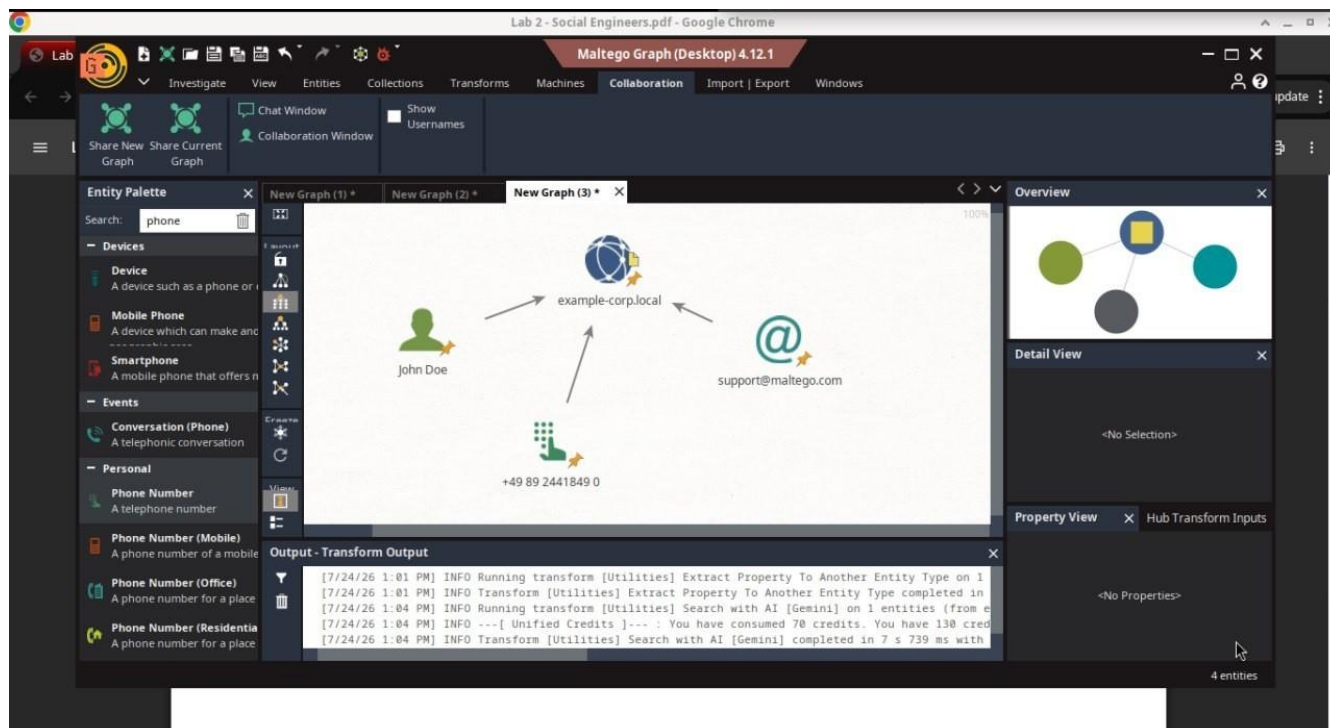


Figure 2: Relationship Graph Showing Connected Entities.

Yes, your instructor specifically asks you to **add notes to the entities with fictional information**. You should include this as a separate step in your report.

Add the following section after **Step 5** (or wherever it fits based on your screenshots).

Step 4: Add Notes to Entities

After creating and connecting the entities, notes were added to each entity to provide additional fictional information and context.

Person Entity

Entity: John Doe

Observation

The note provides descriptive information about the employee, making the graph more informative and easier to understand.

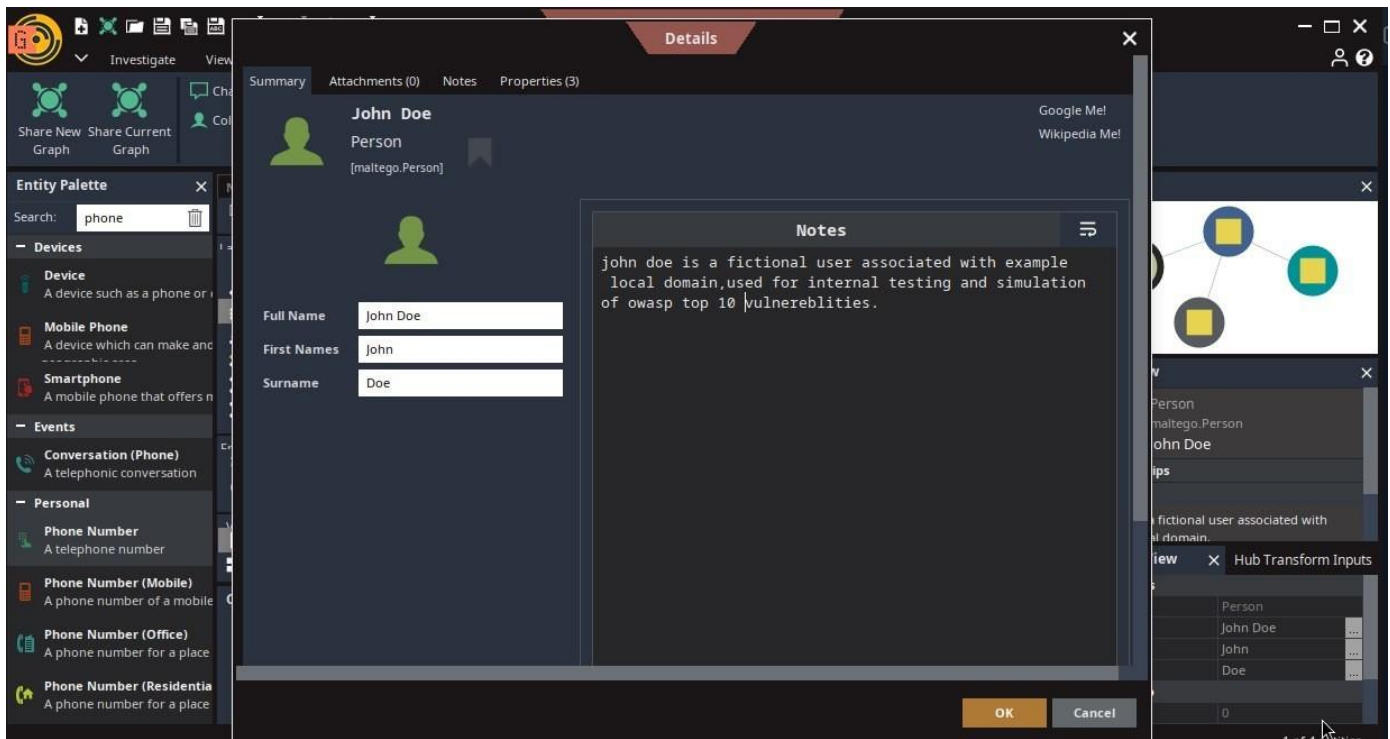


Figure 3: Person Entity with Attached Note.

Email Entity

Entity:

support@maltego.com

Observation

The email entity was documented with fictional information representing the organization's email format.

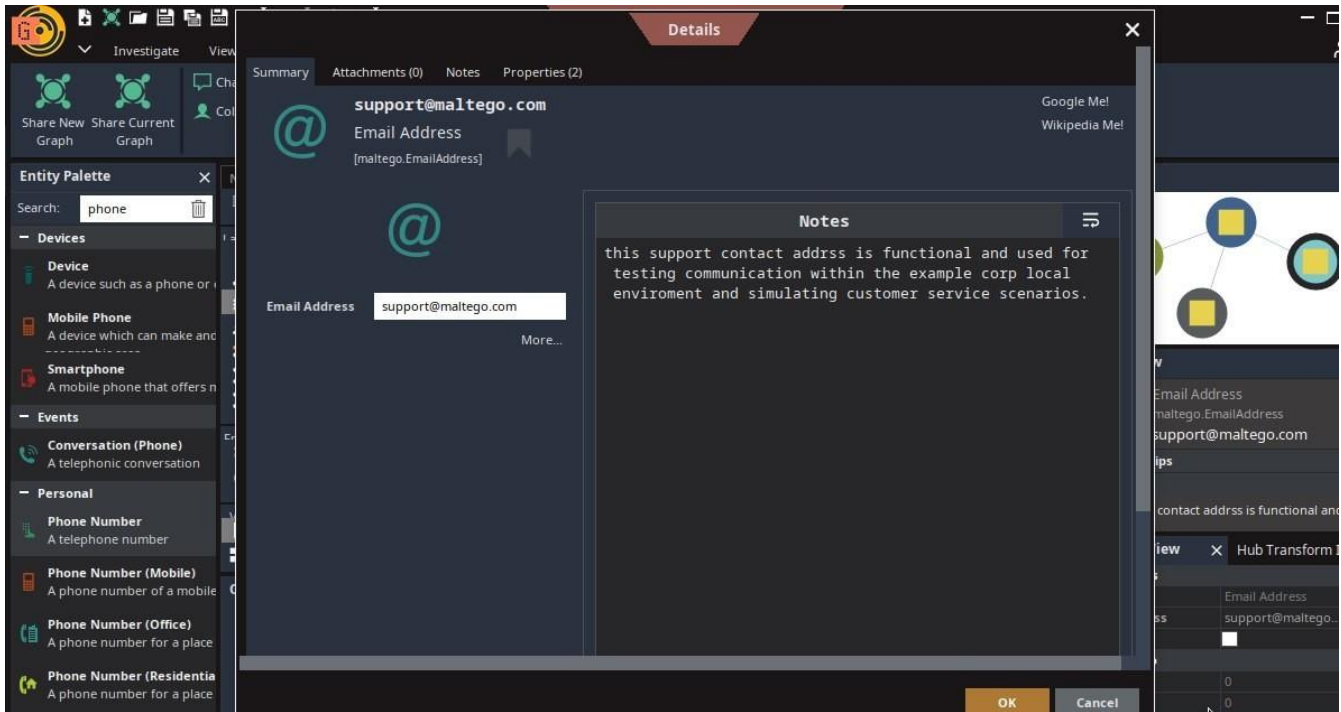


Figure 4: Email Entity with Attached Note.

Phone Number Entity

Observation

The phone number entity was annotated with fictional contact information to simulate publicly available organizational data.

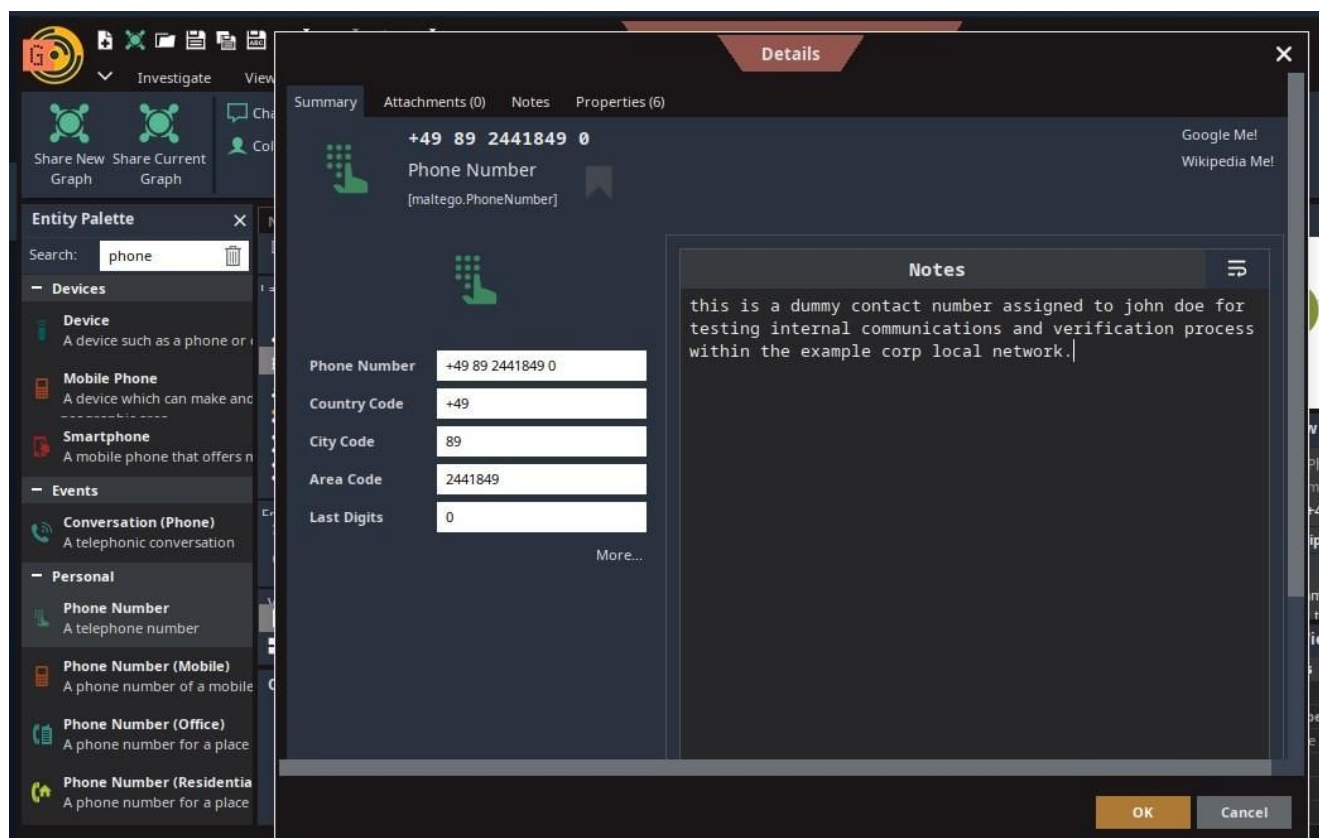


Figure 5: Phone Number Entity with Attached Note.

Domain Entity

Entity

example-corp.local

Observation

The Domain entity was documented with fictional organizational information to clearly identify the simulated environment and distinguish it from real-world domains.

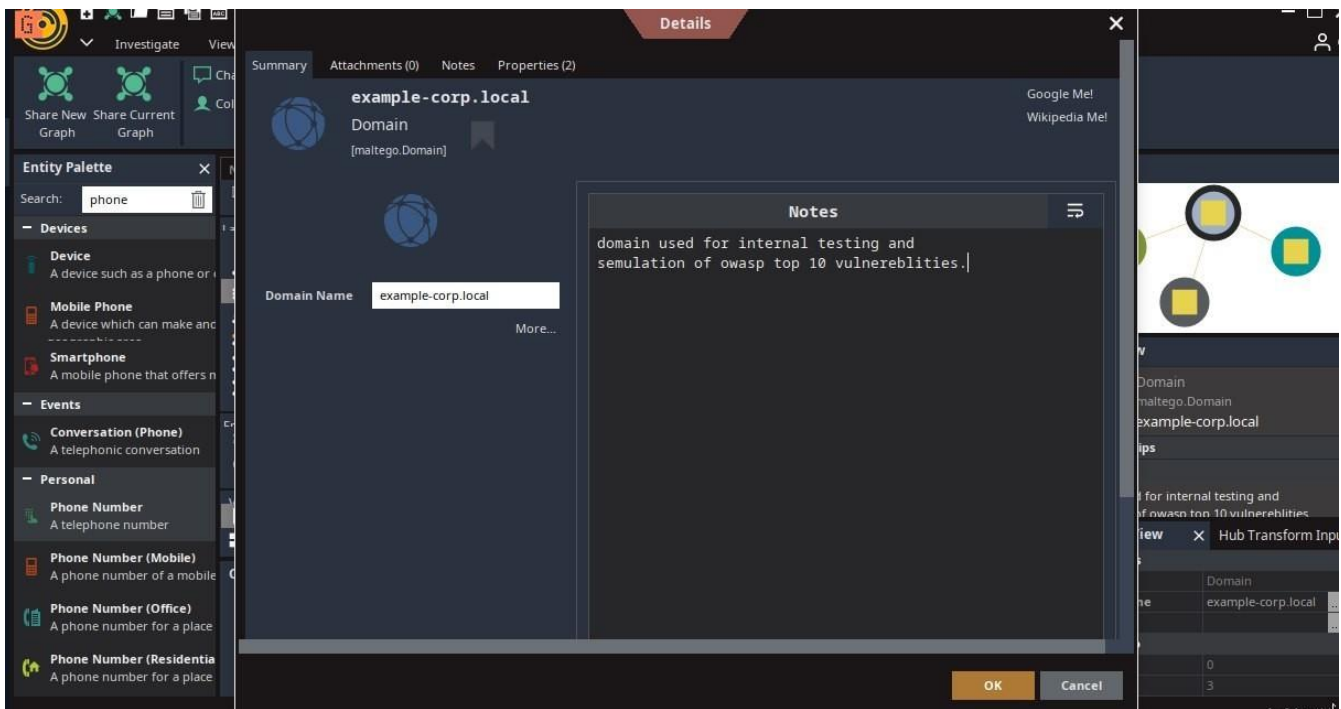


Figure 6: Domain Entity with Attached Note.

Step 5: Relationship Between Entities

The following relationships were manually created:

Observation

The manually connected entities created a clear visualization of how an employee is associated with an organization, email address, and telephone number. This demonstrates Maltego's ability to represent relationships between different types of information in a graphical format.

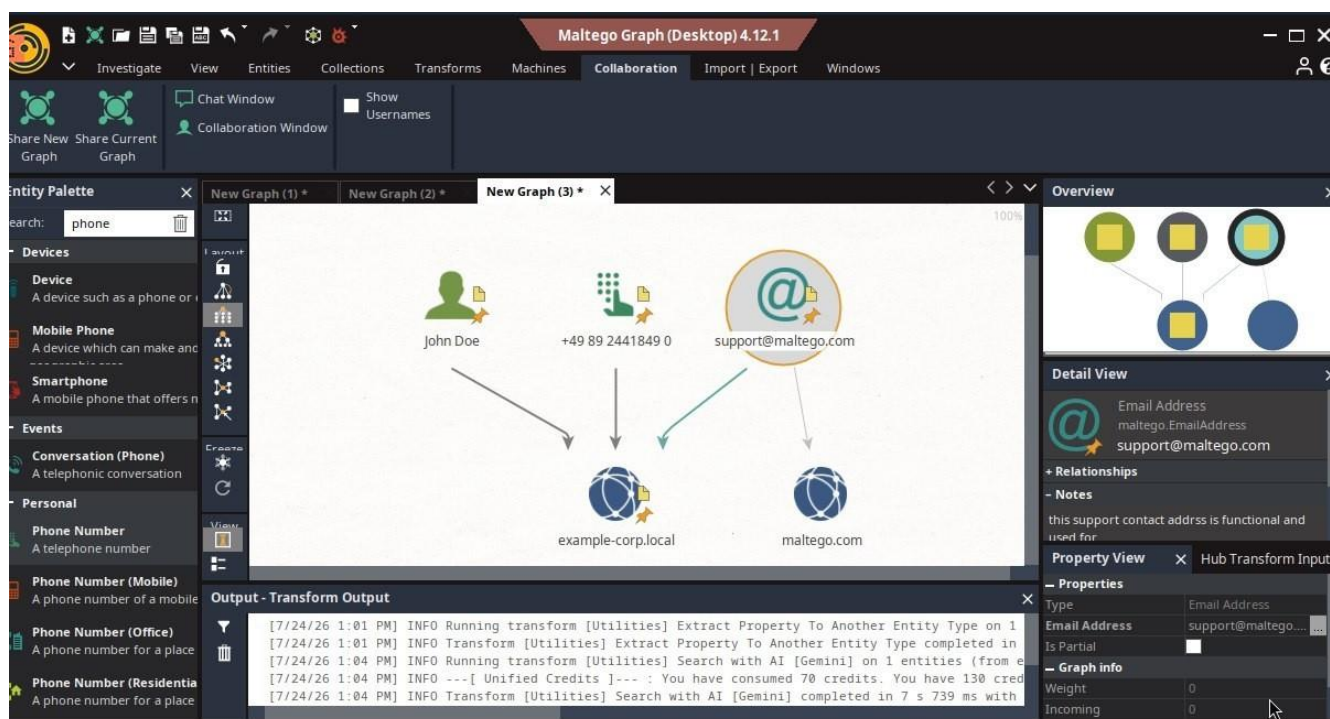


Figure 9: Connected Entities Showing Relationships.

Step 6: Understanding Transform Types

Objective

The objective of this step was to understand how Maltego transforms convert one entity into another by discovering relationships using publicly available information. Transforms automate the OSINT process by expanding an investigation and revealing connections between entities.

1. Email to Domain

Description

The **Email to Domain** transform extracts the domain name from an email address.

Observation

The transform identified the domain associated with the email address, allowing further investigation of the organization's infrastructure.

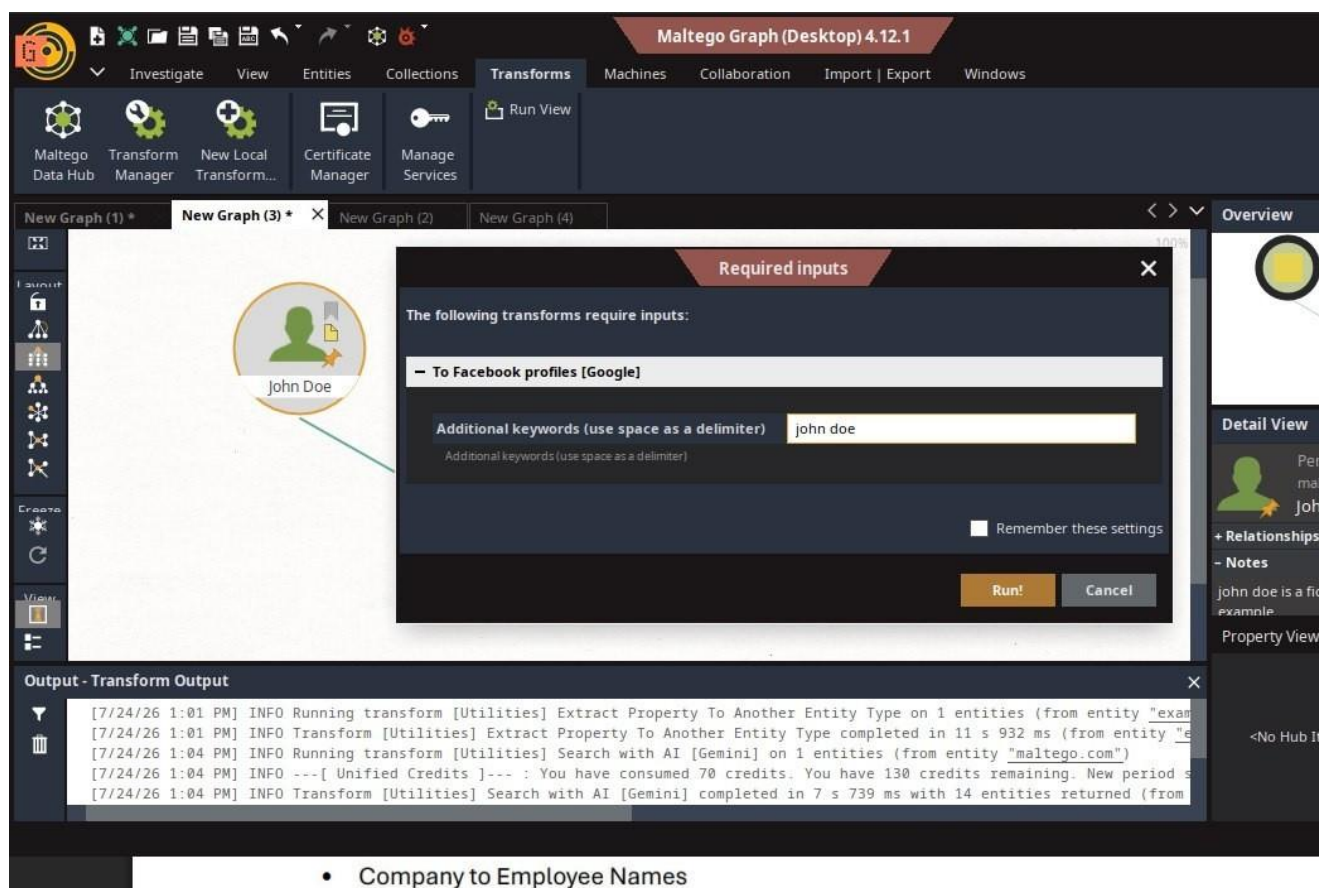
2. Person to Social Media Profiles

Description

The **Person to Social Media Profiles** transform searches for publicly available social media accounts associated with a person's name.

Observation

The transform demonstrated how publicly available information may reveal an individual's online presence and professional or personal profiles.



Have no social media

Figure 10: Person to Social Media Profiles Transform.

3. Domain to IP Address

Description

The **Domain to IP Address** transform resolves a domain name to its associated IP address.

Observation

The transform identified the server associated with the target domain, enabling further investigation of the network infrastructure.

4. Domain to Email Addresses

Description

The **Domain to Email Addresses** transform searches for publicly available email addresses associated with the target domain.

Observation

The transform identified email addresses related to the organization, demonstrating how contact information may be discovered through OSINT techniques.

5. Company to Employee Names

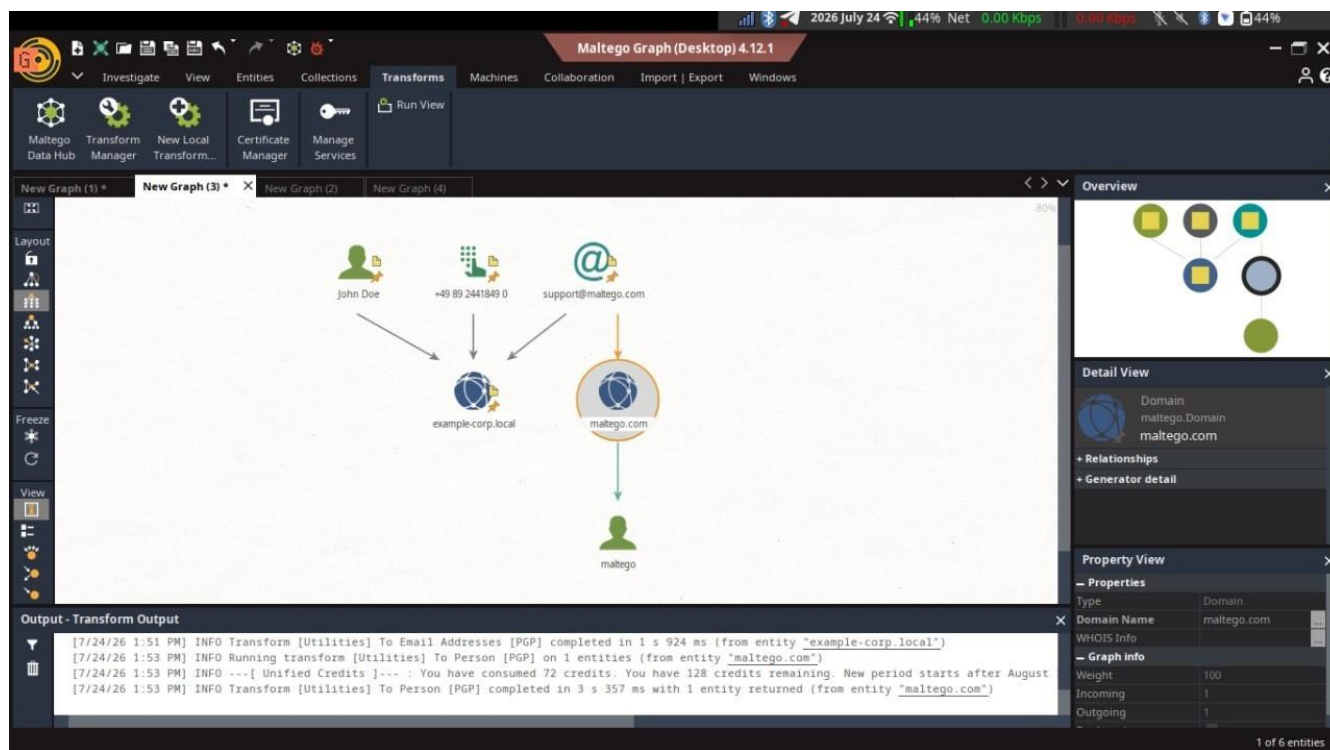
Description

The **Company to Employee Names** transform searches for publicly available information linking an organization to its employees.

Observation

The transform identified employee names associated with the organization, illustrating how publicly available sources can reveal organizational structure and personnel information.

All number 3 up 5 steps are included in this screenshots



Analysis:

- How does visual relationship mapping aid reconnaissance?
- What information sources does Maltego aggregate?
- How could attackers use this information for targeted phishing?

1. How does visual relationship mapping aid reconnaissance?

Visual relationship mapping helps analysts identify connections between entities such as domains, IP addresses, email addresses, and organizations. It simplifies large amounts of OSINT data into an easy-to-understand graph, making it easier to discover relationships, identify infrastructure, and understand a target's digital footprint.

2. What information sources does Maltego aggregate?

Maltego gathers information from multiple OSINT sources, including:

- DNS and WHOIS records
- Domain and IP information
- Search engines
- Social media and public profiles
- Threat intelligence platforms
- Publicly available databases and online services

3. How could attackers use this information for targeted phishing?

Attackers can use OSINT gathered through Maltego to identify employees, email addresses, domains, and organizational relationships. This information enables them to create convincing spear-phishing emails that appear legitimate, increasing the likelihood that victims will trust the message and disclose sensitive information or credentials.